Supplementary Online Content

Scheepers MHMC, Al-Difaie Z, Brandts L, Peeters A, van Grinsven B, Bouvy ND. Diagnostic performance of electronic noses in cancer diagnoses using exhaled breath: a systematic review and meta-analysis. *JAMA Netw Open*. 2022;5(6):e2219372. doi:10.1001/jamanetworkopen.2022.19372

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This supplementary material has been provided by the authors to give readers additional information about their work.

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eTable 1: Full electronic search strategy performed in the PubMed-, and Embase- databases.

#	Search
1.	Lung Diseases/
2.	Gastrointestinal Diseases/
3.	Neoplasms/
4.	lung disease$.ti,ab,kw.
5.	gastrointestinal disease$.ti,ab,kw.
6.	carcinom$.ti,ab,kw.
7.	neoplasm$.ti,ab,kw.
8.	cancer$.ti,ab,kw.
9.	mesothelioma$.ti,ab,kw.
10.	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9
11.	Electronic Nose/
12.	Volatile Organic Compounds/
13.	artificial nose$.ti,ab,kw.
14.	electronic nose$.ti,ab,kw.
15.	e-nose$.ti,ab,kw.
16.	eNose$.ti,ab,kw.
17.	VOC$.ti,ab,kw.
18.	volatile organic compound$.ti,ab,kw.
19.	breath anal$.ti,ab,kw.
20.	machine olfact$.ti,ab,kw.
21.	artificial olfact$.ti,ab,kw.
22.	aeonose$.ti,ab,kw.
23.	cyranose$.ti,ab,kw.
24.	11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23
25.	Diagnosis/
26.	"Sensitivity and Specificity"/
27.	diagnosis$.ti,ab,kw.
28.	diagnostic$.ti,ab,kw.
29.	sensitivity$.ti,ab,kw.
30.	specificity$.ti,ab,kw.
31.	ROC curve$.ti,ab,kw.
32.	ROC analys$.ti,ab,kw.
33.	receiver operating characteristic$.ti,ab,kw.
34.	25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33
35.	10 and 24 and 34
36.	remove duplicates from 35

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eTable 2: Modified QUADAS-2 assessment tool

		QUADAS-2 (modified)
RISK of BIAS	**Patient selection**	Were sampled patients representative of the intended population?
		Did the study include patients as well as healthy controls?
		Did the study avoid inappropriate exclusions?
		Could the selection of patients have introduced bias?
	Index test	Was the index test and interpretation of data performed in standardized and reproducible fashion?
		If a threshold was used, was it pre-specified?
		Validation of results performed (internal or external)? Did they validate their results on a blinded sample?
		Could the conduct or interpretation of the index test have introduced bias?
	Reference standard	Is the reference standard likely to correctly classify the target condition?
		Were the reference standard results interpreted without knowledge of the results of the index test?
		Is the reference standard likely to correctly classify the control group?
		Could the reference standard, its conduct, or its interpretation have introduced bias?
	Flow and timing	Was there an appropriate interval between index test and reference standard?
		Did all patients receive a reference standard?
		Were index test and reference standard performed prior to therapeutic intervention?
		Did all patients receive the same reference standard?
		Were measures taken to limit exogenous influences such as food intake and smoking etc?
		Were all patients included in the analysis?
		Could the patient flow and application of the tests have introduced bias?
APPLICABILITY	**Patient selection**	Are there concerns that the included patients and setting do not match the review question?
	Index test	Are there concerns that the authors have not demonstrated suitable reproducibility and sensitivity of the chosen index test?
	Reference standard	Are there concerns that the target condition as defined by the reference standard does not match the question**?**

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eTable 3: The Rational Clinical Examination: Evidence-Based Clinical Diagnosis >TABLE: Levels of Evidence

Levels of Evidence[a]

Level of Evidence	Grade	Definition
1	A	Independent blinded comparison of sign or symptom results with a criterion standard of diagnosis among a large number of consecutive patients suspected of having the target condition
2	B	Independent blinded comparison of sign or symptom with a criterion standard of diagnosis among a small number of consecutive patients suspected of having the target condition
3	C	Independent blinded comparison of sign or symptom with a criterion standard of diagnosis among nonconsecutive patients suspected of having the target condition
4	C	Nonindependent comparison of sign or symptom with a criterion standard of diagnosis among samples of patients who obviously have the target condition plus, perhaps, normal individuals
5	C	Nonindependent comparison of sign or symptom with a standard of uncertain validity

[a]Modified from Holleman and Simel.

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eTable 4: Examples of reported confounding factors and measures to reduce influences.

Factor	Measure	References
Food and beverages	Fasting before breath sampling	(1-7, 9, 14, 22, 24-43)
Smoking	Cessation of smoking before breath sampling	(1, 2, 4, 6, 7, 9, 14, 24-26, 28, 30-33, 35, 36, 38, 40-45)
Expiratory compartment	Alveolar or mixed exhaled breath.	(1-8, 12, 14, 28, 30, 31, 34, 38, 41, 43, 45, 46)
Sample collection	Exhaled breath can be collected in breath sampling bags, Tenax tubes or directly exposed to electronic nose sensors.	Tedlar: (1, 3, 5, 9, 25-27, 30, 31, 34, 36-40, 43, 45-47) Mylar**:** (1, 6-8, 12, 48) Gassampler: **(**2, 4, 14) Other**:** (22), (41)(16, 17)(29, 49) Direct: (11, 13, 15, 18-21, 23, 32, 33, 42, 44, 50, 51)
Exercise	Exercise may affect the composition of VOCs found in exhaled breath. A period of resting before breath sampling might be beneficial.	Resting**:** (22, 25, 33, 40, 43)
Oral VOCs	VOCs produced by the oral microbiome or lingering VOCs from diet might influence VOCs in exhaled breath. Mouth rinsing	(22-27)
Perfume aromatics	Cessation of use of perfume, deodorant or pungent foods.	Perfume/aromatics/ mouth hygiene: (3), (25, 28-33)

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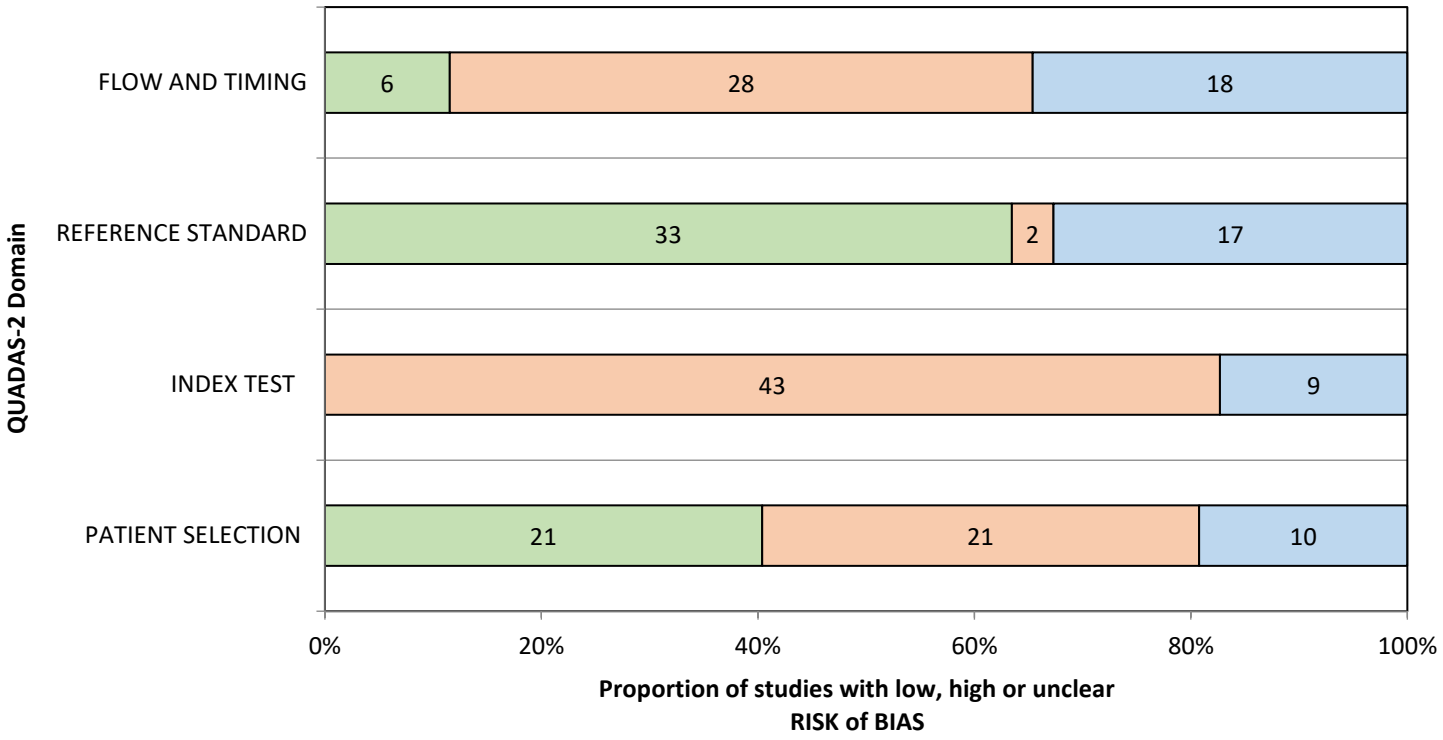
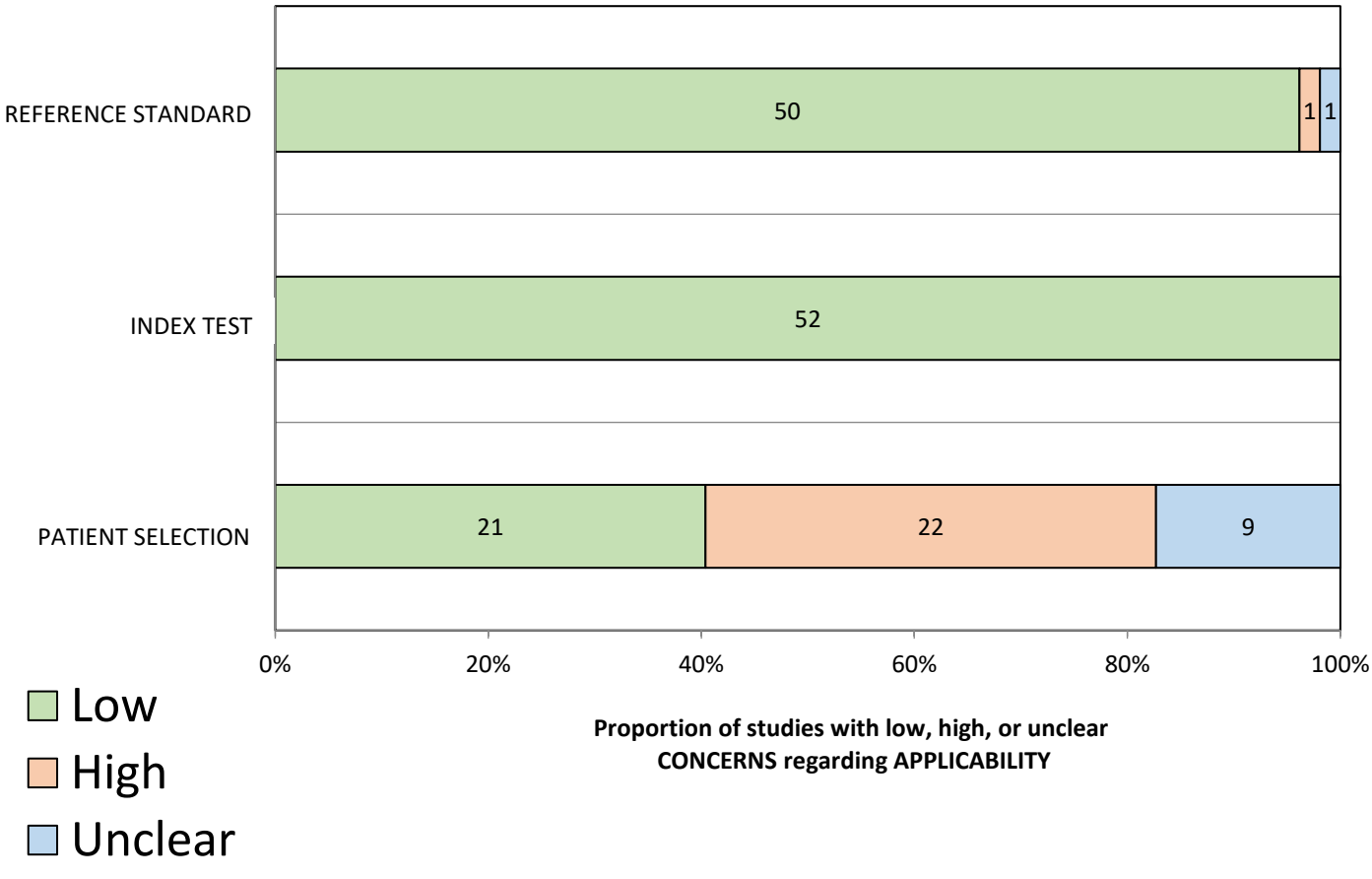
eTable 5: Quality assessment using QUADAS-2 tool: Summary and separate outcome of risk of bias and concerns regarding applicability for included studies.





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eTable 6: Quality assessment using QUADAS-2 tool: Outcomes of risk of bias and applicability concerns assessment for individual studies

Study	Risk of Bias				Applicability concerns		
	Patient Selection	Index test	Reference standard	Flow and timing	Patient selection	Index test	Reference standard
Altomare et al, 41 2016	☺	☹	☺	?	☺	☺	☺
Amal et al,6 2014	☹	☹	☺	?	?	☺	☺
Amal et al, 44 2015	?	?	☺	?	?	☺	☺
Amal et al, 45 2016	☺	?	☺	?	☺	☺	☺
Barash et al,64 2015	☺	?	☺	?	☺	☺	☺
Broza et al, 46 2012	☺	☹	☺	☹	☺	☺	☺
Capuano et al, 65 2015	☹	☹	?	☹	☹	☺	☺
Chapman et al, 32 2012	?	☹	☺	☹	☺	☺	☺
Chen et al, 47 2021	☹	?	☹	☹	☹	☺	?
Chen Q et al, 48 2020	☺	☹	☺	?	☺	☺	☺
de Kort et al, 23 2018	☺	☹	?	☹	☹	☺	☺
de Kort et al, 22 2020	☺	☹	☺	☹	☹	☺	☺
de Vries et al,66 2015	☹	☹	☺	☹	☹	☺	☺
Di Natale et al,67 2003	☹	☹	?	☹	☹	☺	☺
Diaz de Leon-Martinez et al, 33 2020	☺	☹	☺	?	☺	☺	☺
Dragonieri et al, 68 2009	☹	☹	☺	?	☹	☺	☺
Dragonieri et al, 16 2012	☹	☹	☺	☹	☹	☺	☺
Gasparri et al, 49 2016	☹	☹	☺	☹	☹	☺	☺

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Gruber et al, 50 2014	☺	?	☺	☺	☹	☺	☺
Hakim et al,51 2011	?	☹	?	☹	☺	☺	☺
Herman-Saffar et al, 34 2018	☹	☹	?	☹	?	☺	☺
Huang et al, 35 2018	☹	☹	☺	☺	☹	☺	☺
Hubers et al, 36 2014	?	☹	☺	☹	?	☺	☺
Kononov et al, 52 2019	☺	☹	?	?	☹	☺	☺
Krauss et al, 24 2020	?	☹	☺	☹	☹	☺	☺
Lamote et al, 37 2017	☺	☹	?	?	☺	☺	☺
Leja et al,53 2021	☺	?	☺	?	☹	☺	☺
Leunis et al, 54 2014	☹	☹	?	☹	☺	☺	☺
Li et al, 55 2017	☹	?	?	☹	☺	☺	☺
Li et al, 69 2020	☹	?	☺	☹	☹	☺	☺
Liu et al, 56 2021	☺	☹	☺	☹	☺	☺	☺
Machado et al, 17 2005	☺	☹	☺	☹	☹	☺	☺
Marzorati et al, 57 2019	☹	☹	?	☹	?	☺	☺
Mohamed et al, 25 2021	?	☹	☺	☹	?	☺	☺
Mohamed et al, 42 2019	☹	☹	☺	☺	☺	☺	☺
Peled et al,58 2012	☺	☹	☺	☹	☺	☺	☺
Raspaglie si et al, 43 2020	☺	☹	☺	☺	☹	☺	☺
Rocco et al, 59 2016	?	☹	?	?	?	☺	☺
Schuerma ns et al, 26 2018	☺	☹	☺	?	?	☺	☺

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Shehada et al, 60 2016	☹	☹	?	☹	☹	☺	☺
Shlomi et al, 61 2017	☺	☹	☺	☺	☺	☺	☺
Steenhuis et al, 27 2020	☺	☹	☺	?	☺	☺	☺
Tan et al, 62 2016	☺	☹	☺	☹	☺	☺	☺
Tirzite et al,38 2017	?	☹	?	?	☹	☺	☺
Tirzite et al, 39 2019	?	☹	?	?	☹	☺	☺
van de Goor et al, 28 2018	☹	☹	?	☹	☹	☺	☺
van de Goor et al, 18 2019	☺	☹	?	?	☺	☺	☺
van de Goor et al, 29 2020	☹	☹	?	☹	☹	☺	☺
van Keulen et al, 30 2020	?	☹	☺	?	☺	☺	☺
Waltman, 31 2020	☹	☹	☺	☹	?	☺	☺
Xu et al,63 2013	☹	?	☺	☺	☺	☺	☺
Yang et al, 40 2021	☹	☹	☹	☹	☺	☺	☹

☺ low risk ☹ high risk ? unclear risk

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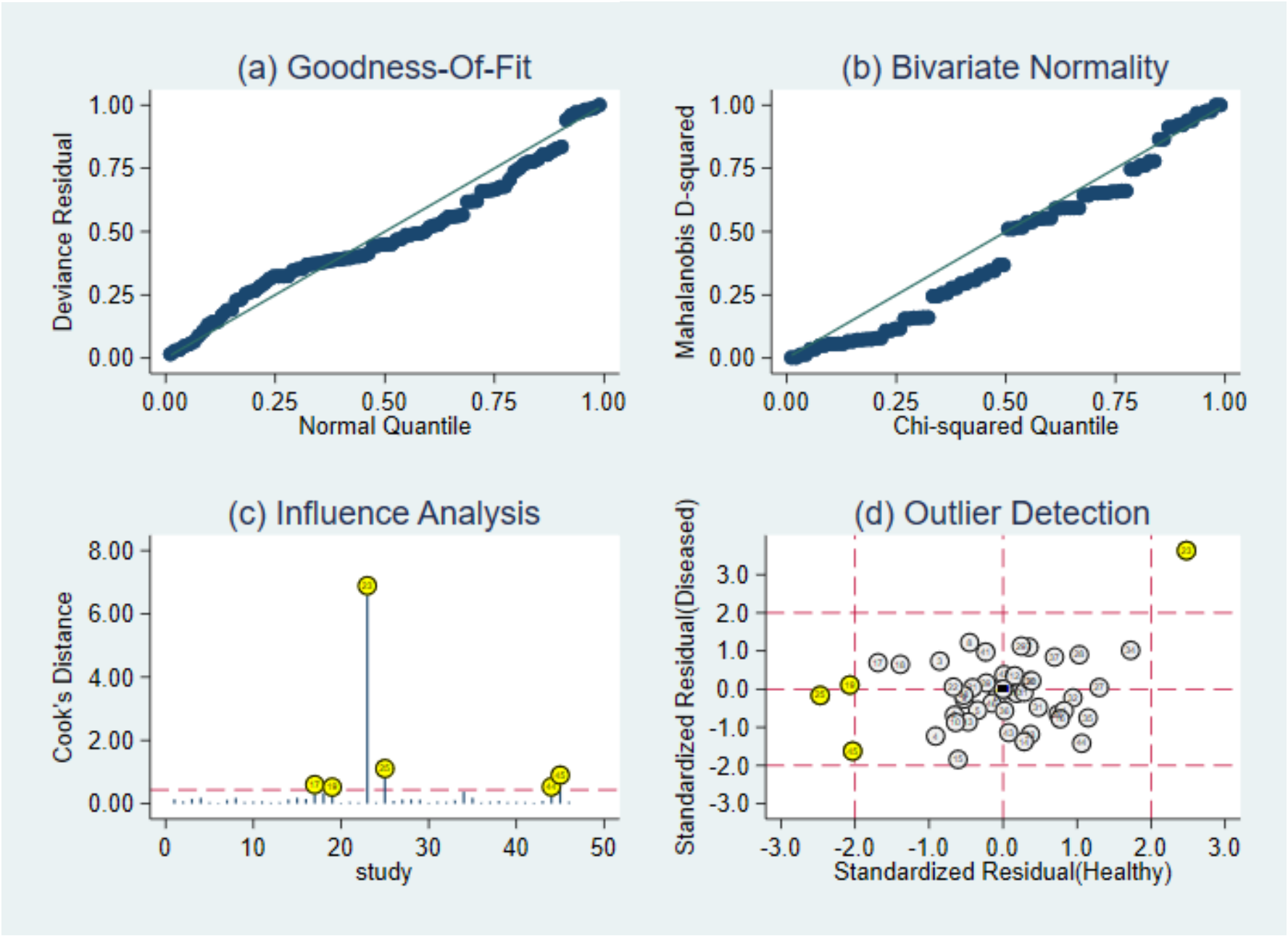
eFigure 1: Outlier and Influence analysis of all e-nose studies.



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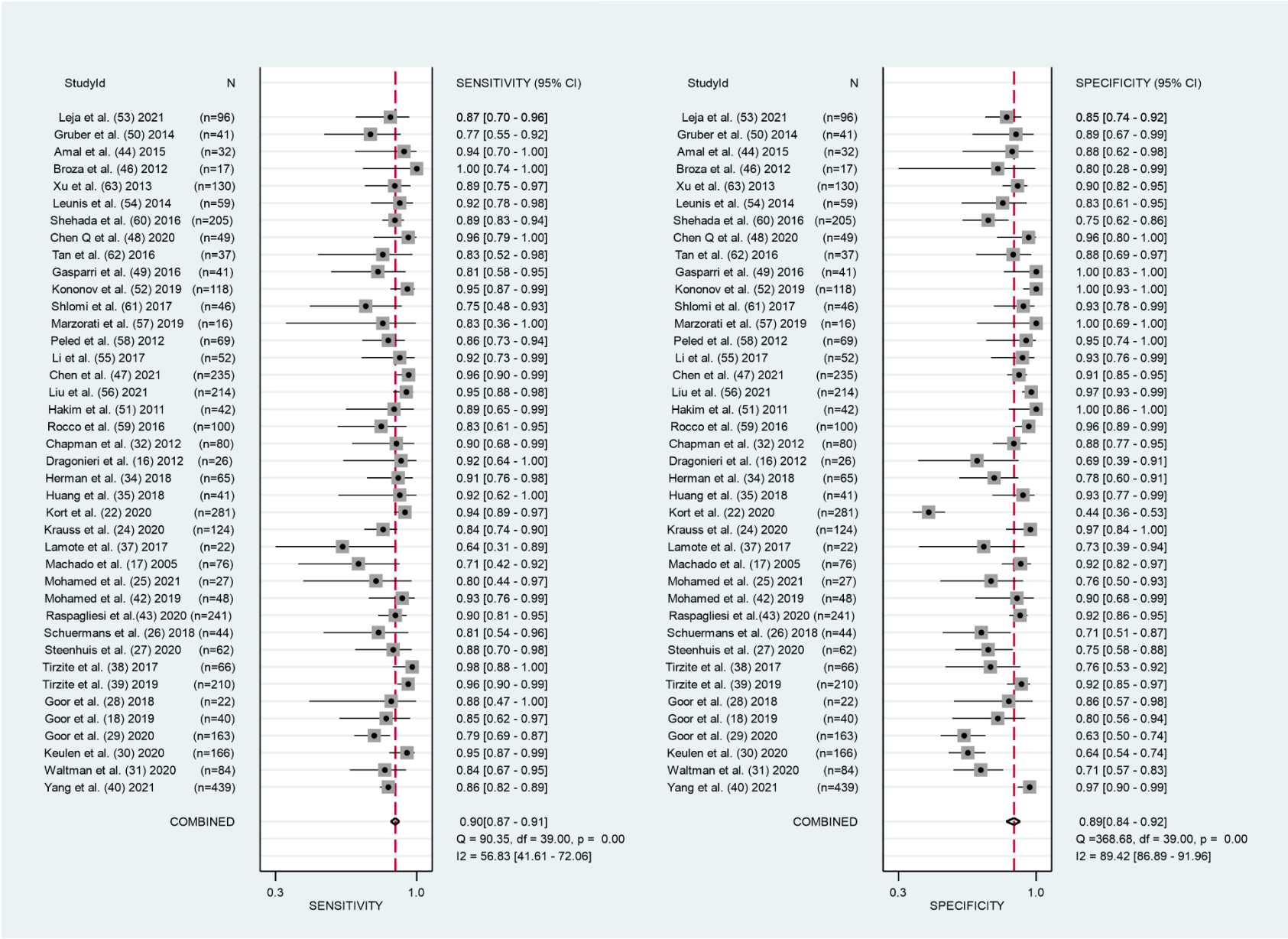
eFigure 2: Pooled analysis of all e-nose studies after exclusion of outliers



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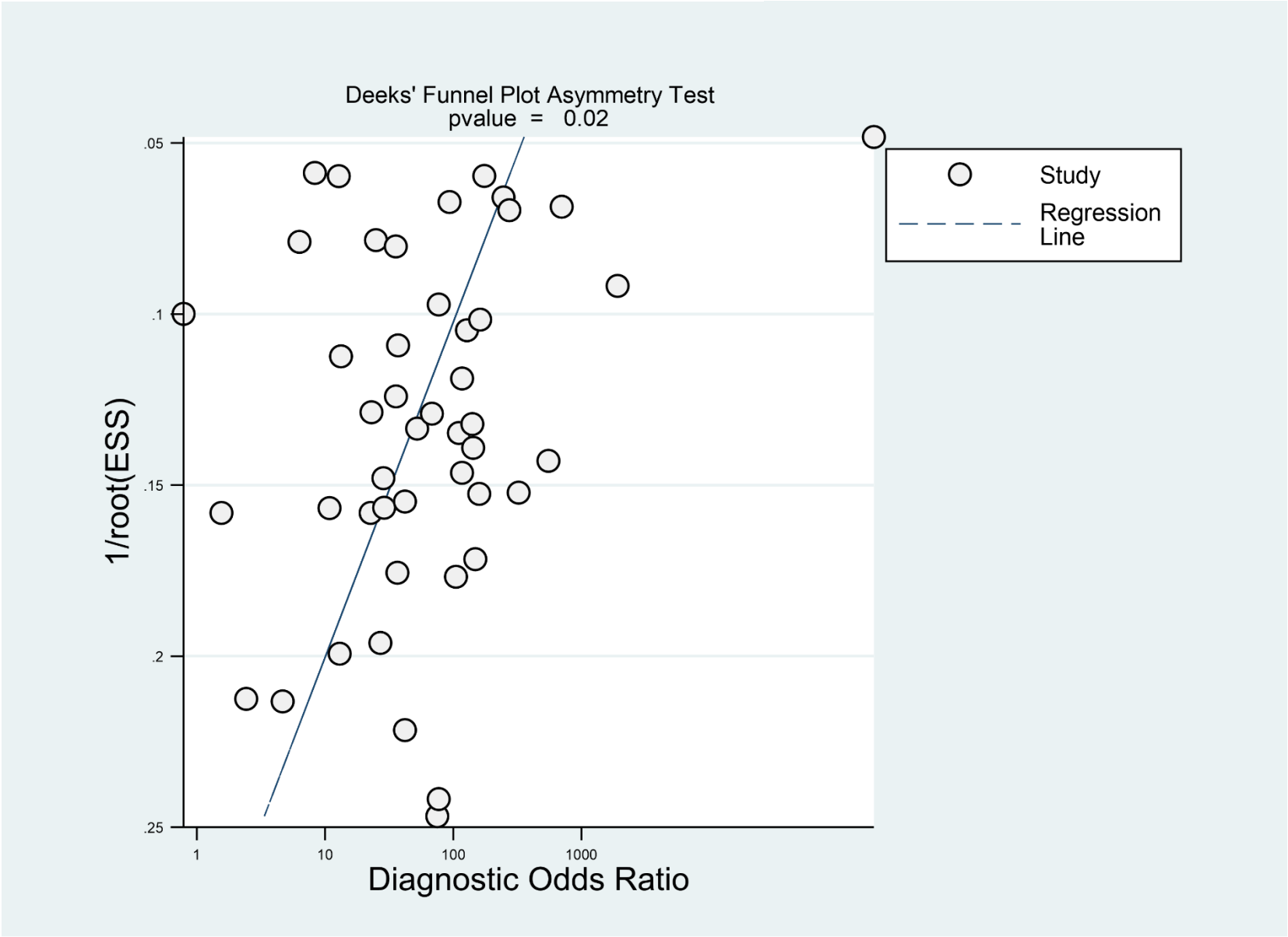
eFigure 3: Publication bias analysis for all e-nose studies.



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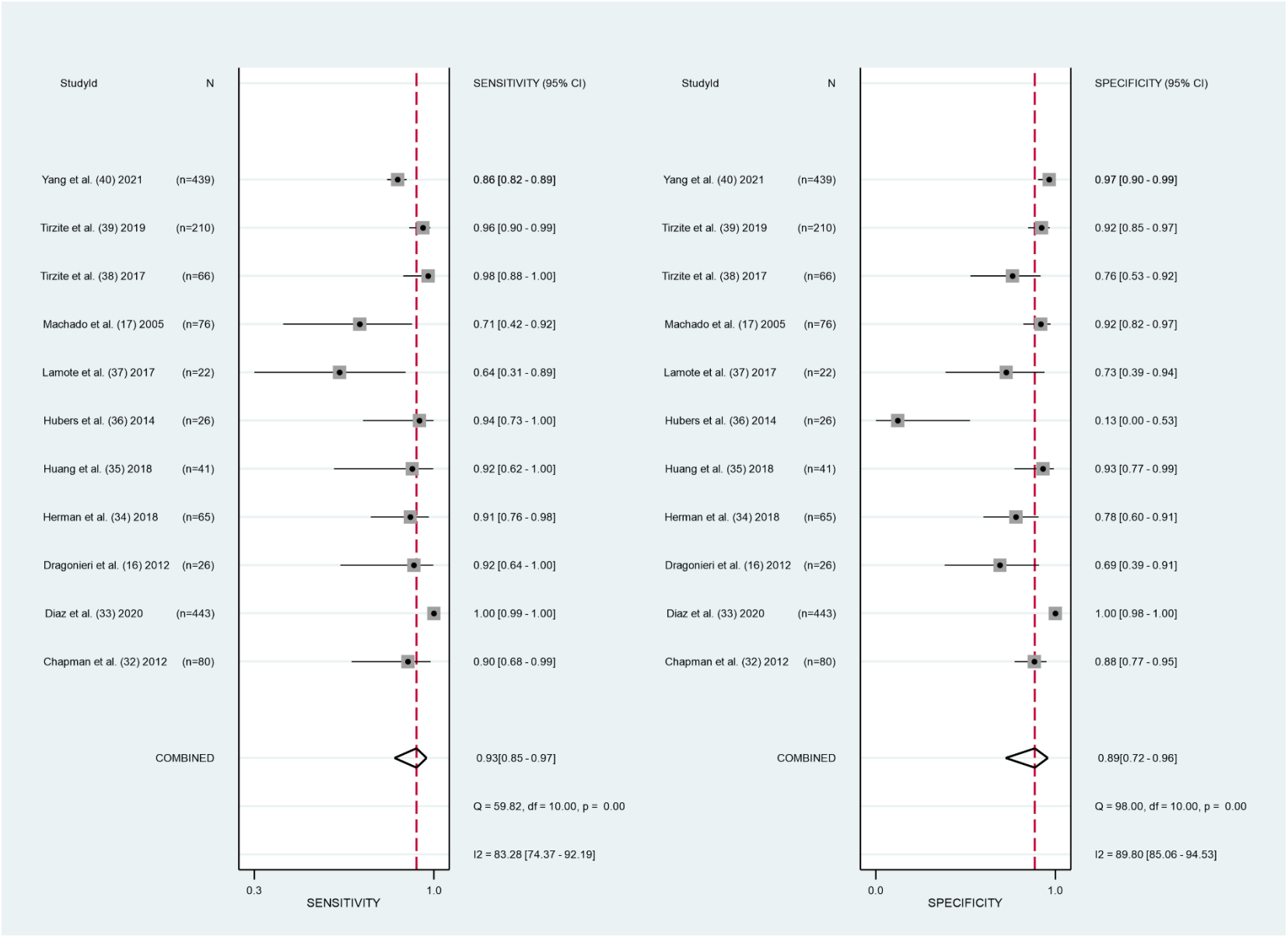
eFigure 4: Pooled analysis of all Cyranose320 studies



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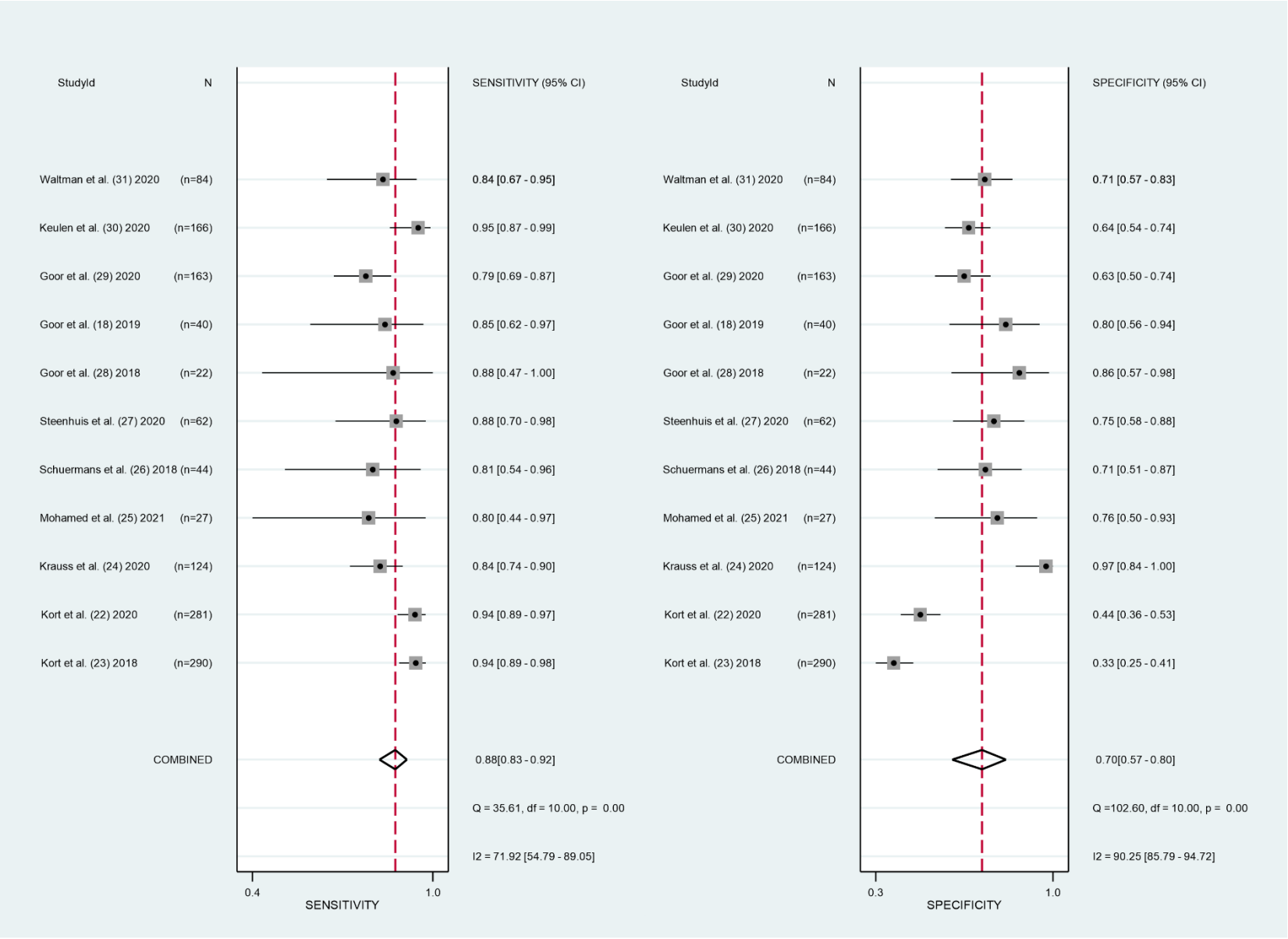
eFigure 5: Pooled analysis of all Aeonose studies



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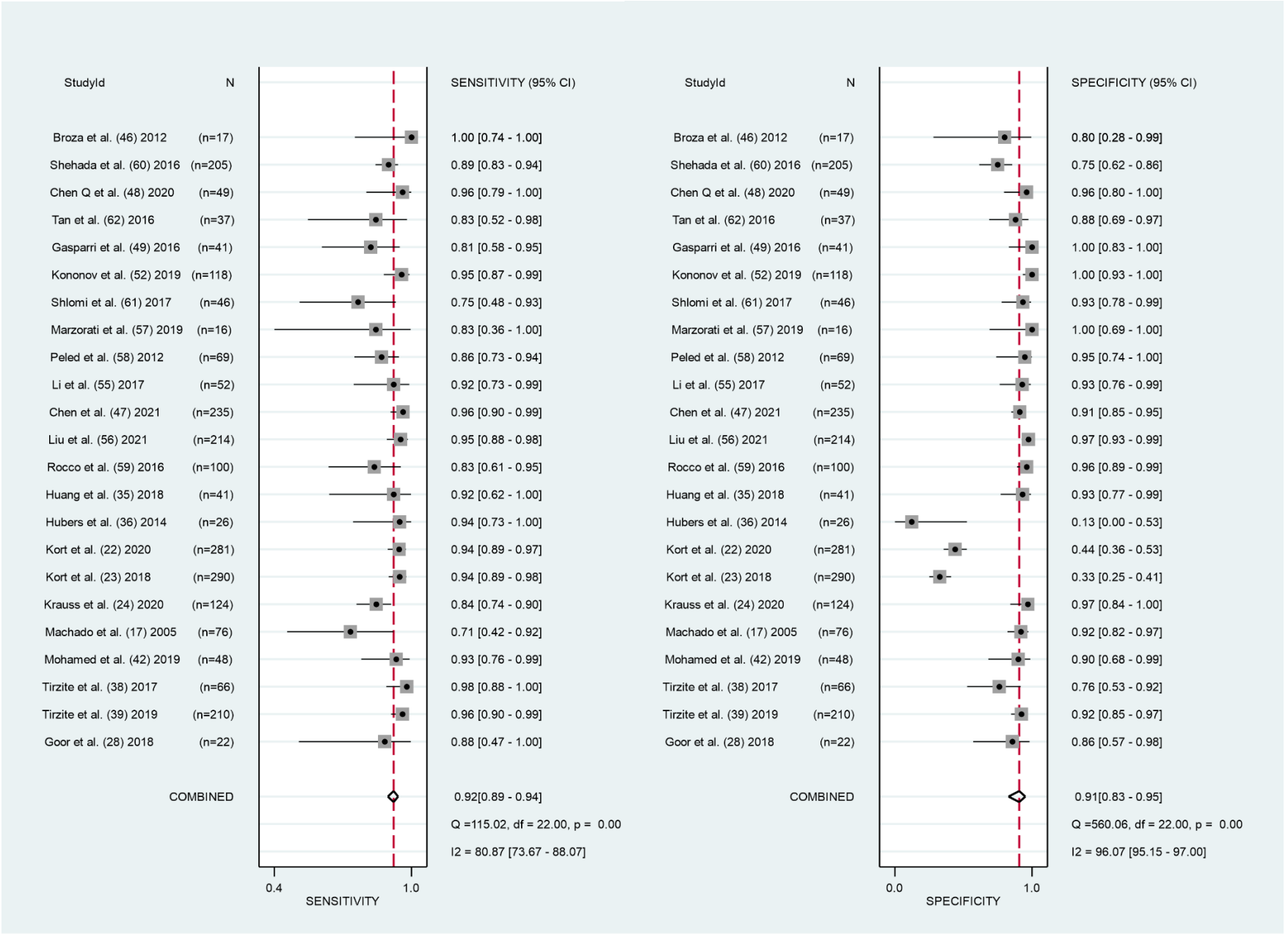
eFigure 6: Pooled analysis of all lung cancer studies



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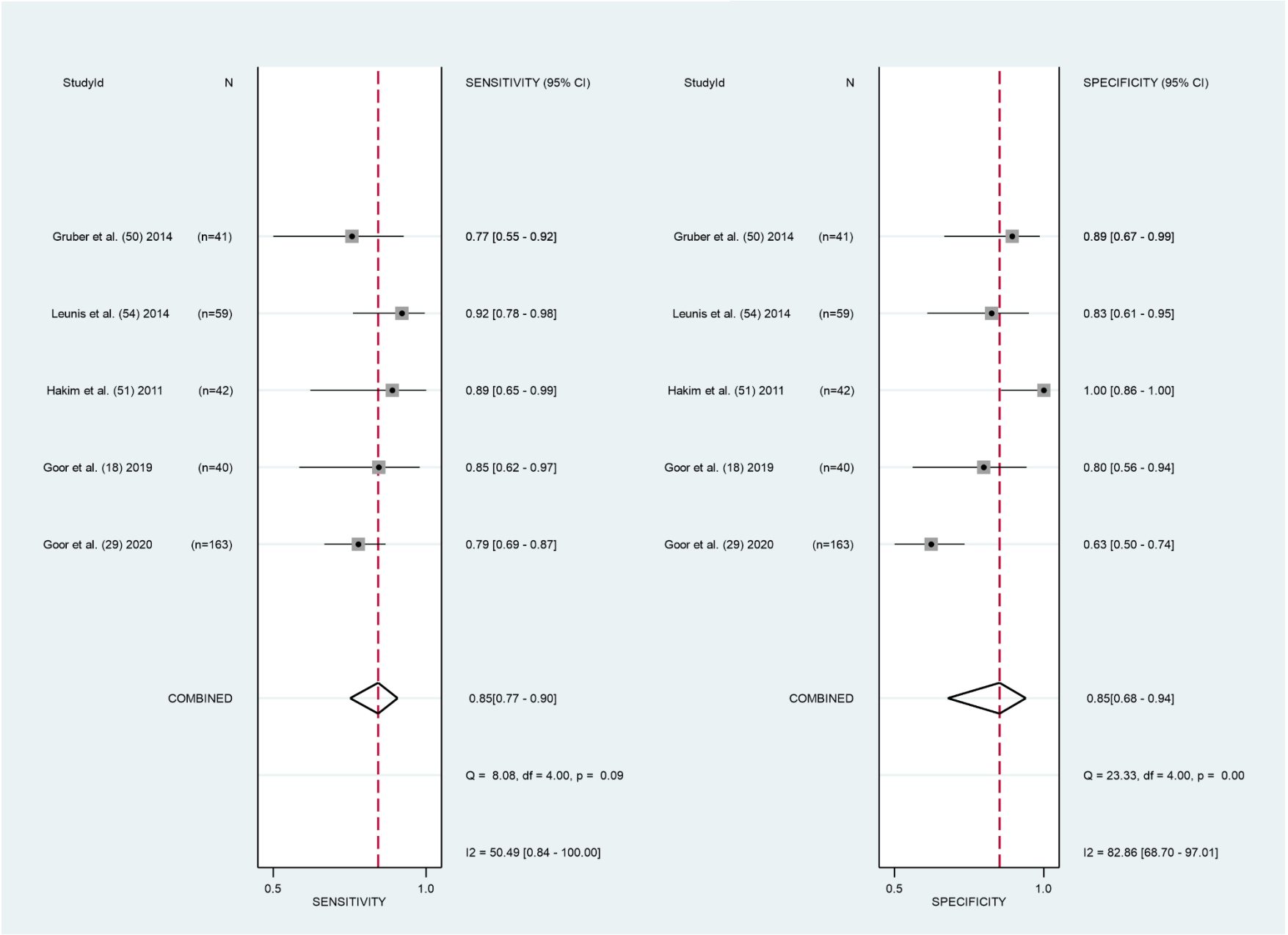
eFigure 7: Pooled analysis of all head and neck cancer studies



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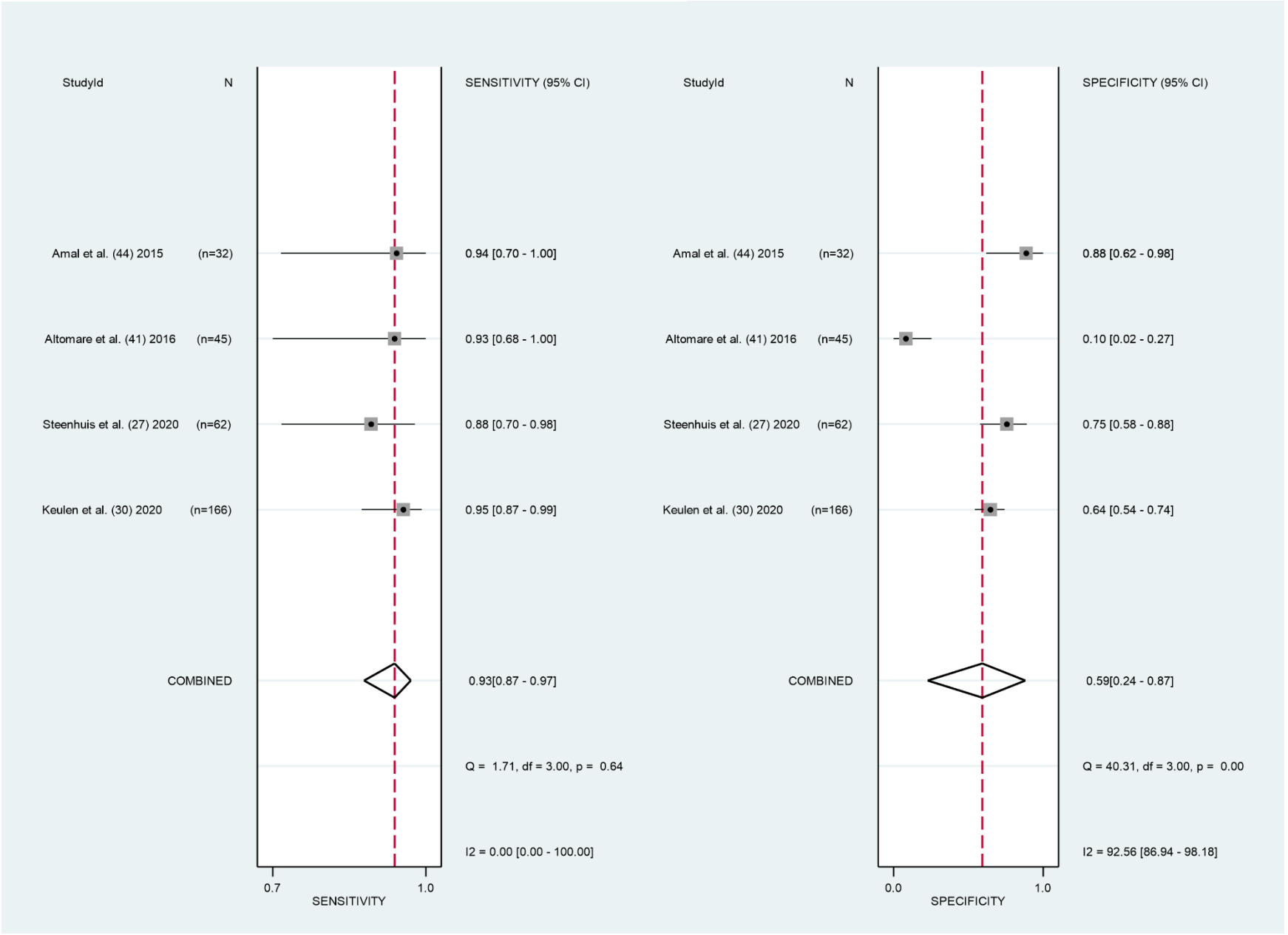
eFigure 8: Pooled analysis of all colorectal cancer studies



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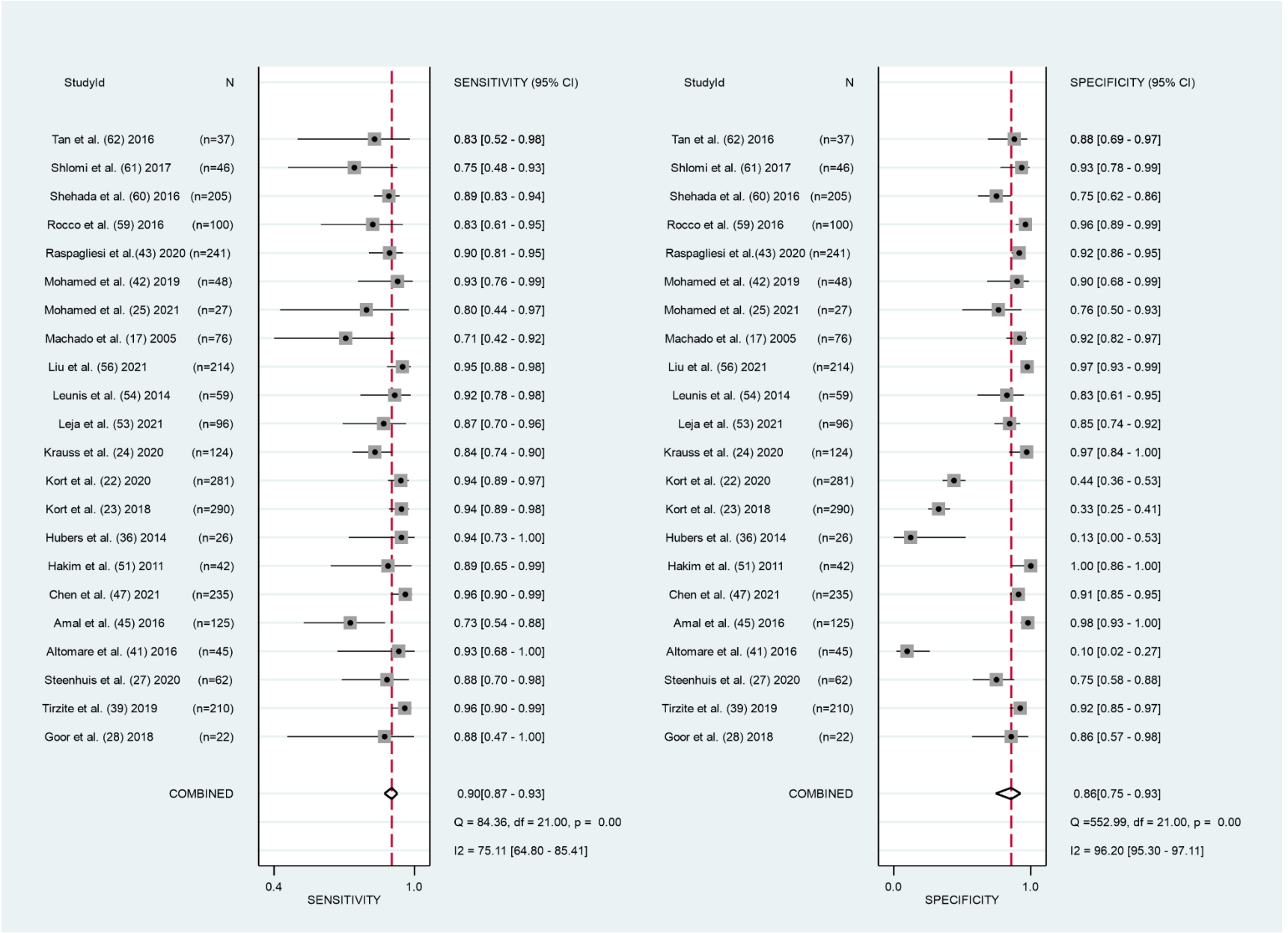
eFigure 9: Pooled analysis of all advanced cancer stage studies



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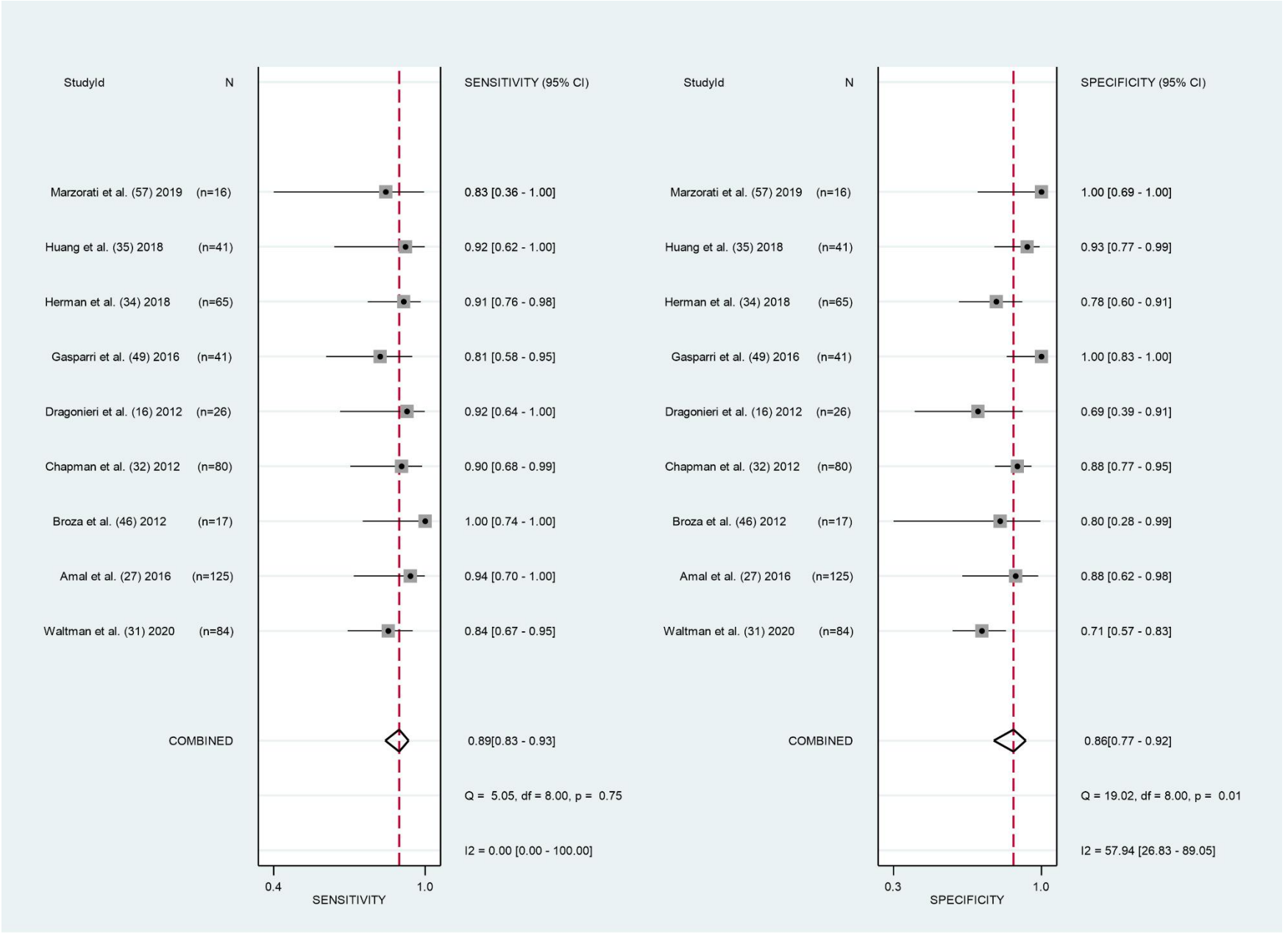
eFigure 10: Pooled analysis of all early cancer stage studies



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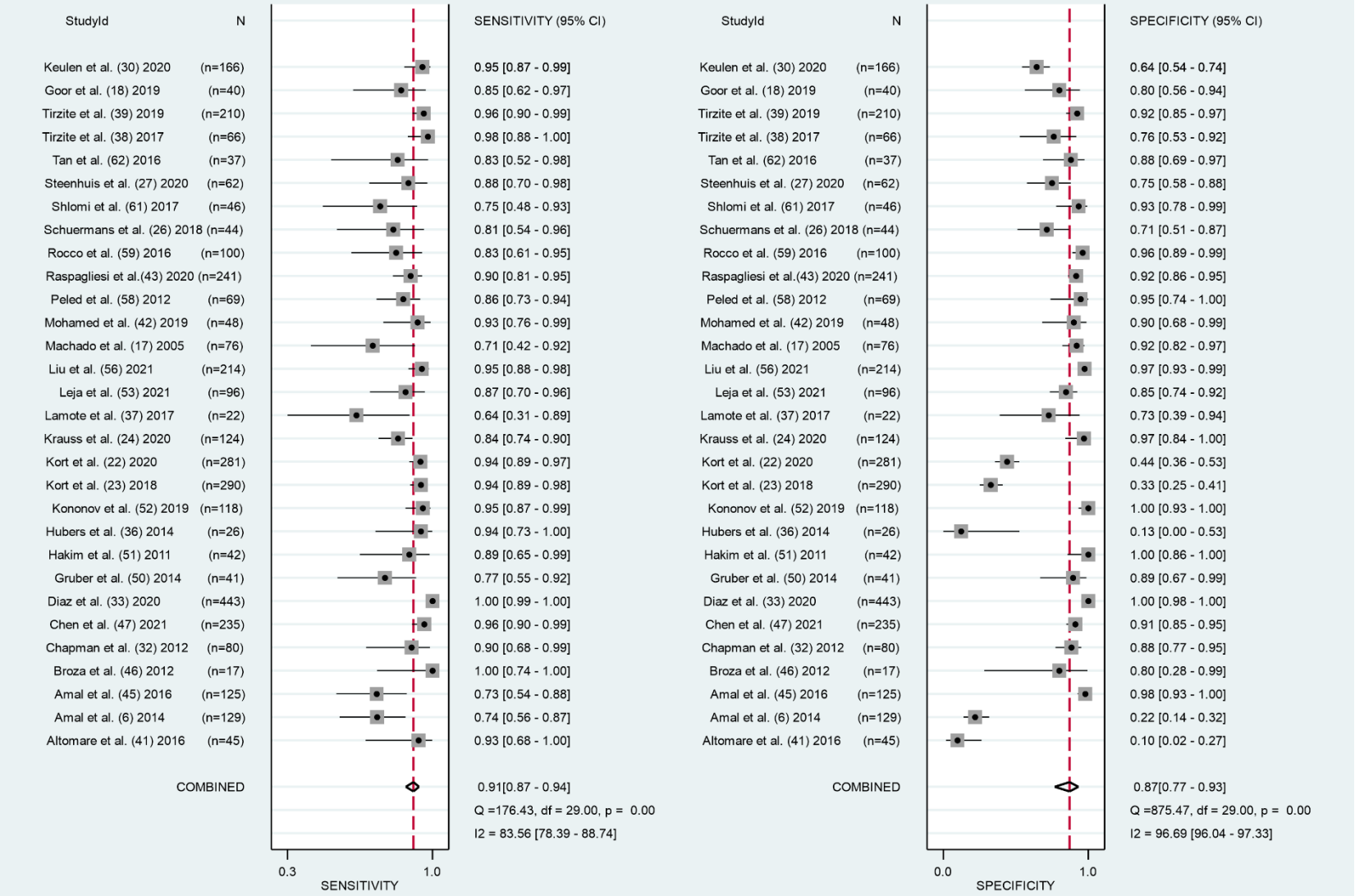
eFigure 11: Pooled analysis of all studies after exclusion of studies with a high risk of bias on the Patient Selection domain of the QUADAS-2 tool



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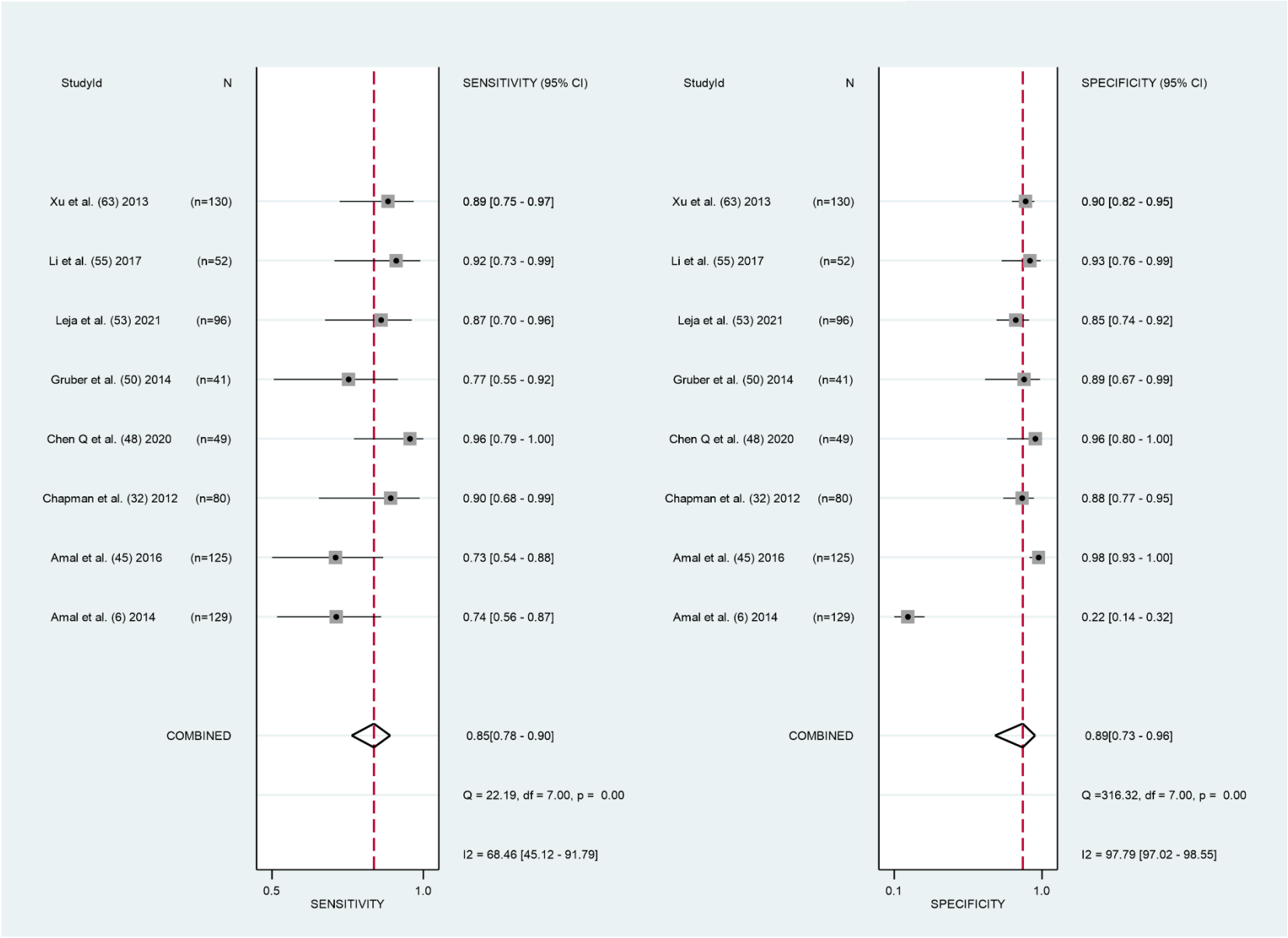
eFigure 12: Pooled analysis of all studies after exclusion of studies with a high risk of bias on the Index Test domain of the QUADAS-2 tool



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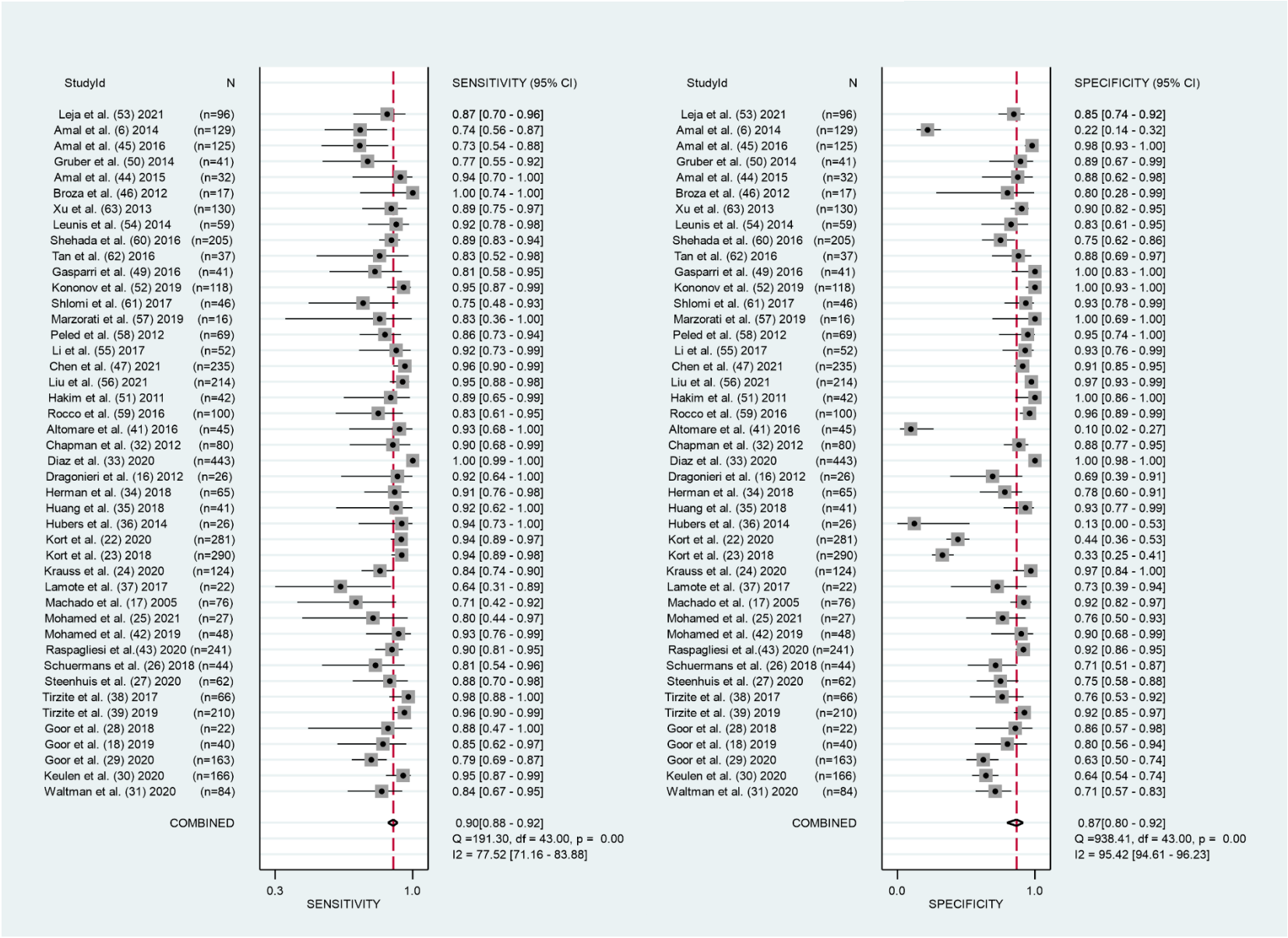
eFigure 13: Pooled analysis of all studies after exclusion of studies with a high risk of bias on the Reference Standard domain of the QUADAS-2 tool



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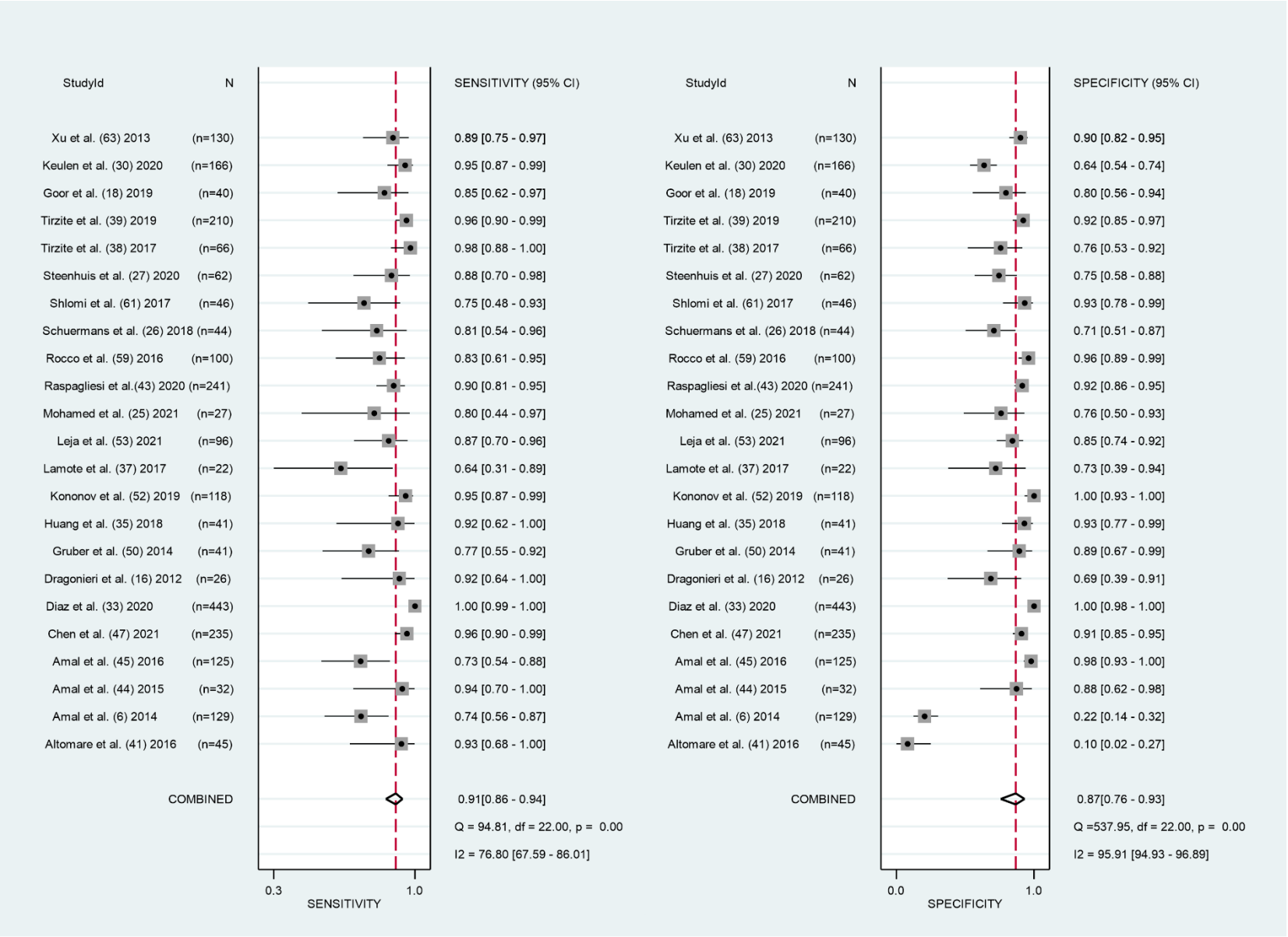
eFigure 14: Pooled analysis of all studies after exclusion of studies with a high risk of bias on the Flow and Timing domain of the QUADAS-2 tool



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